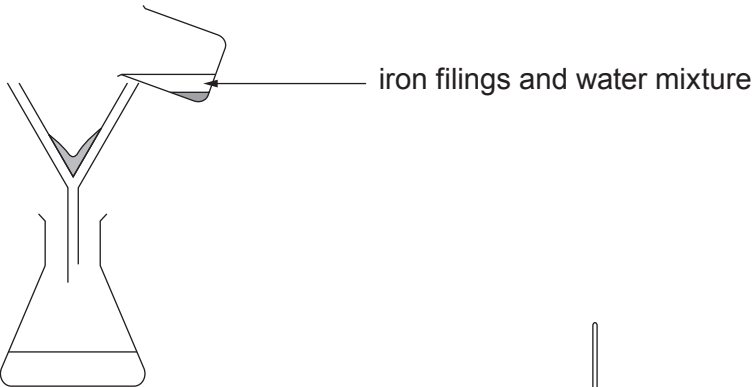


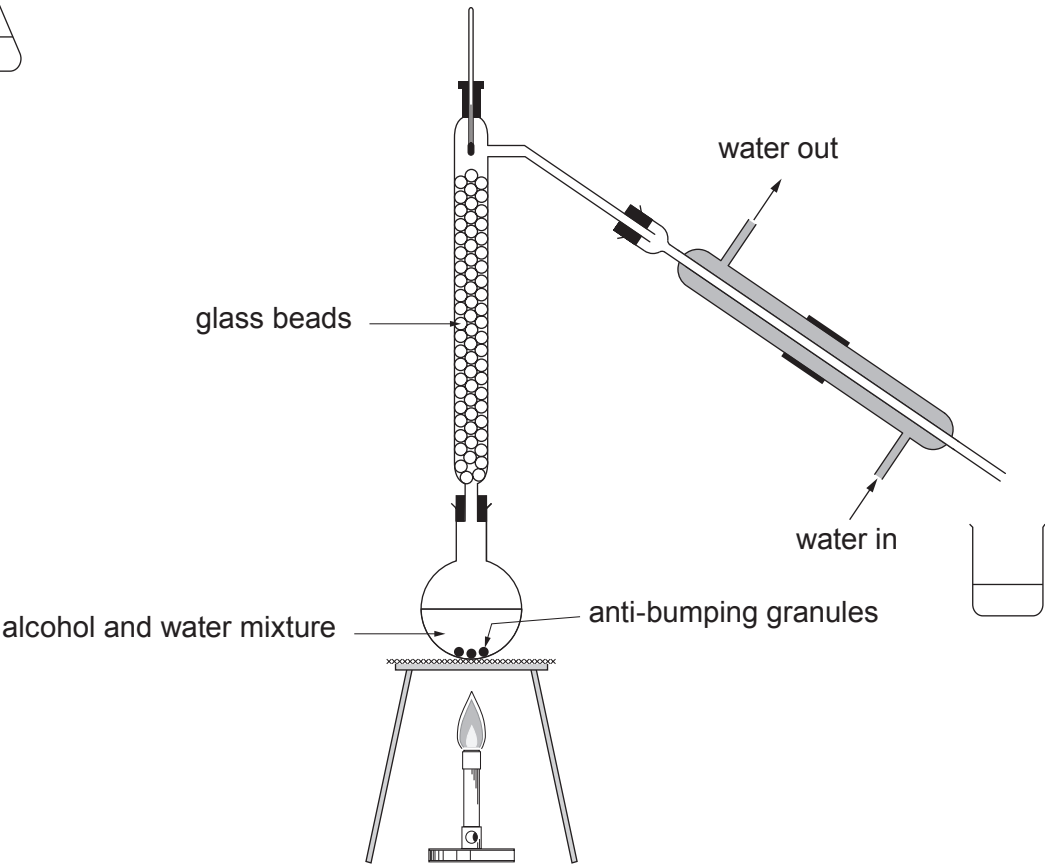
Answer **all** questions.

1. (a) The diagrams show three methods used to separate mixtures.

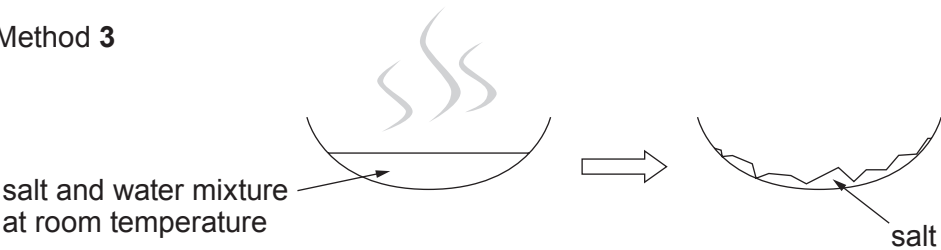
Method 1



Method 2



Method 3





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- (i) Underline the property of iron filings which allows method **1** to be used to separate the mixture. [1]

magnetic insoluble in water silvery grey colour

high melting point more dense than water

- (ii) Tick (✓) the box that shows the reason why method **2** can be used to separate alcohol and water. [1]

alcohol and water both boil when heated ☐

alcohol and water have different boiling points ☐

alcohol and water have the same boiling point ☐

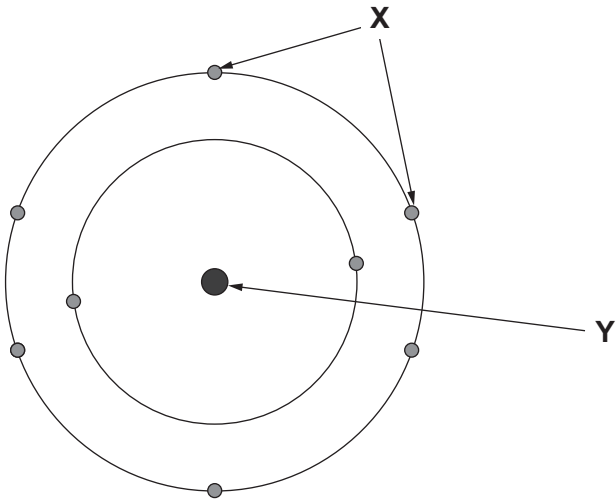
alcohol and water are both liquids ☐

- (iii) Name the process of removing water in method **3**. [1]
-

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Answer all questions.

1. (a) The following diagram shows the structure of an atom.



electrons	protons	shells	neutrons	nucleus
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Complete the following sentences using words from the box. [3]

The particles labelled **X** are called

The part of the atom labelled **Y** is called the

It contains particles called and

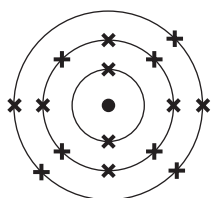
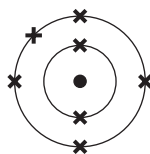
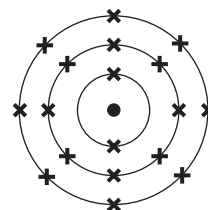
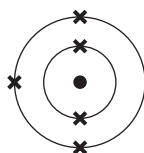
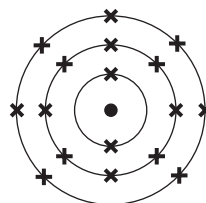
(b) Draw a line from each particle to the charge of that particle. [2]

proton	0
neutron	+1
electron	-1



(c) The diagrams below show five different atoms labelled **A-E**.

These letters are **not** the chemical symbols for the elements.

**A****B****C****D****E**

Use the letters **A-E** to complete the following sentences.

- (i) The atoms found in Group 5 are and [1]
- (ii) The atom with an atomic number of 7 is [1]
- (iii) The atom that is an inert gas is [1]
- (iv) The atoms found in Period 2 are and [1]



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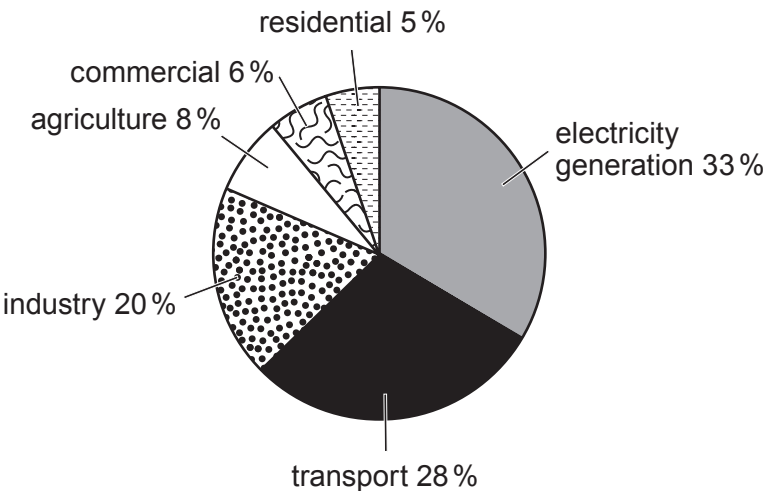


Figure 3

- (a) Draw a trend line on **Figure 1**. [1]
- (b) Suggest why there was a large rise in the amount of carbon dioxide in the atmosphere after around 1950. [2]
-
-
-
-
- (c) Give **one** way in which we are trying to reduce carbon dioxide emissions. [1]
-
-
- (d) Some people say that changes in carbon dioxide levels are **not** responsible for global warming. Tick (✓) the statement that supports this opinion. [1]

Most carbon dioxide is produced by electricity generation

☐

Between 1900 and 2010 there was a massive increase in industry

☐

The mean global temperature remained constant between 1950 and 1980

☐

The average number of cars per home has increased steadily since the 1980s

☐

The use of energy efficient appliances has increased since 2000

☐

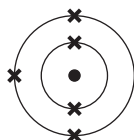
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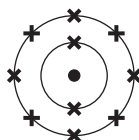
Answer all questions.

1. (a) The diagrams show the atoms of four elements, **A-D**.

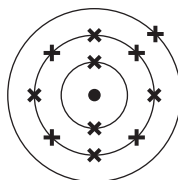
These letters are **not** the chemical symbols for the elements.



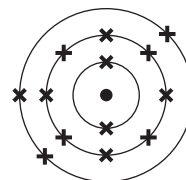
A



B



C



D

- (i) Use the letters **A-D** to complete the following sentences.

Two atoms found in the same group are and

The atom found in Group 1 is

[2]

- (ii) I Give the electronic structure of atom **B**.

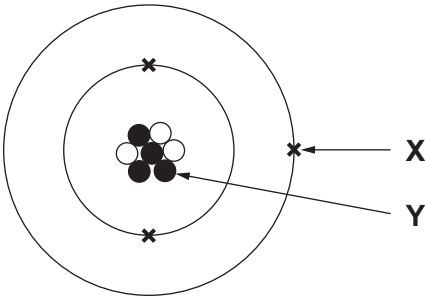
[1]

II Give the atomic number of atom **B**.

[1]



(b) The following diagram shows an atom of lithium.



(i) Complete the following sentences. [2]

Particle **X** is an

Particle **Y** found in the nucleus is a

(ii) Use the diagram to explain why lithium has a mass number of 7. [1]

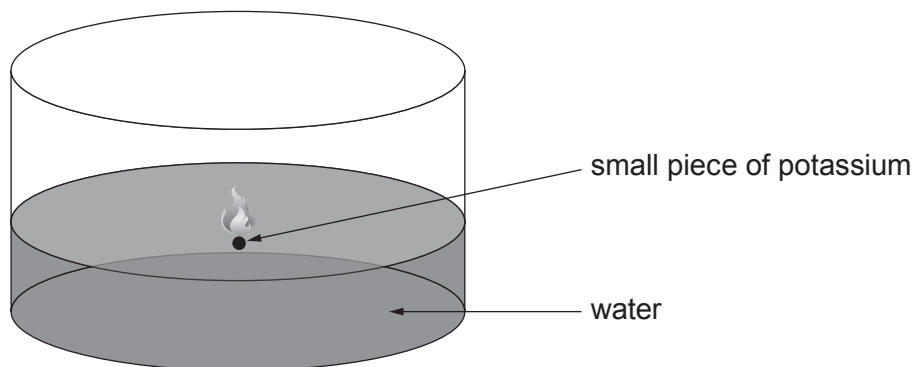
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Answer **all** questions.

1. (a) Potassium reacts with water to produce potassium hydroxide and hydrogen gas.



(i) Complete the table. Tick (✓) **one** box in **each** of the last two rows. [2]

Substance	Formula	Element	Compound
potassium	K	✓	
water	H ₂ O		✓
potassium hydroxide	KOH		
hydrogen	H ₂		

(ii) During the reaction, potassium burns with a **lilac** flame.
Give **two** other observations that you would expect to make.
Choose your answers from the box. [2]

orange crystals form

fizzing

potassium sinks

potassium floats

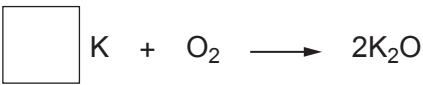
white precipitate forms

Observation 1

Observation 2



(b) Potassium also reacts with oxygen to produce potassium oxide.



Choose a number from the box to balance the equation for this reaction. [1]

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(c) Most Group 1 and Group 2 metals can be identified by the colour seen in a flame test.

Draw **one** line from each metal to the colour seen.

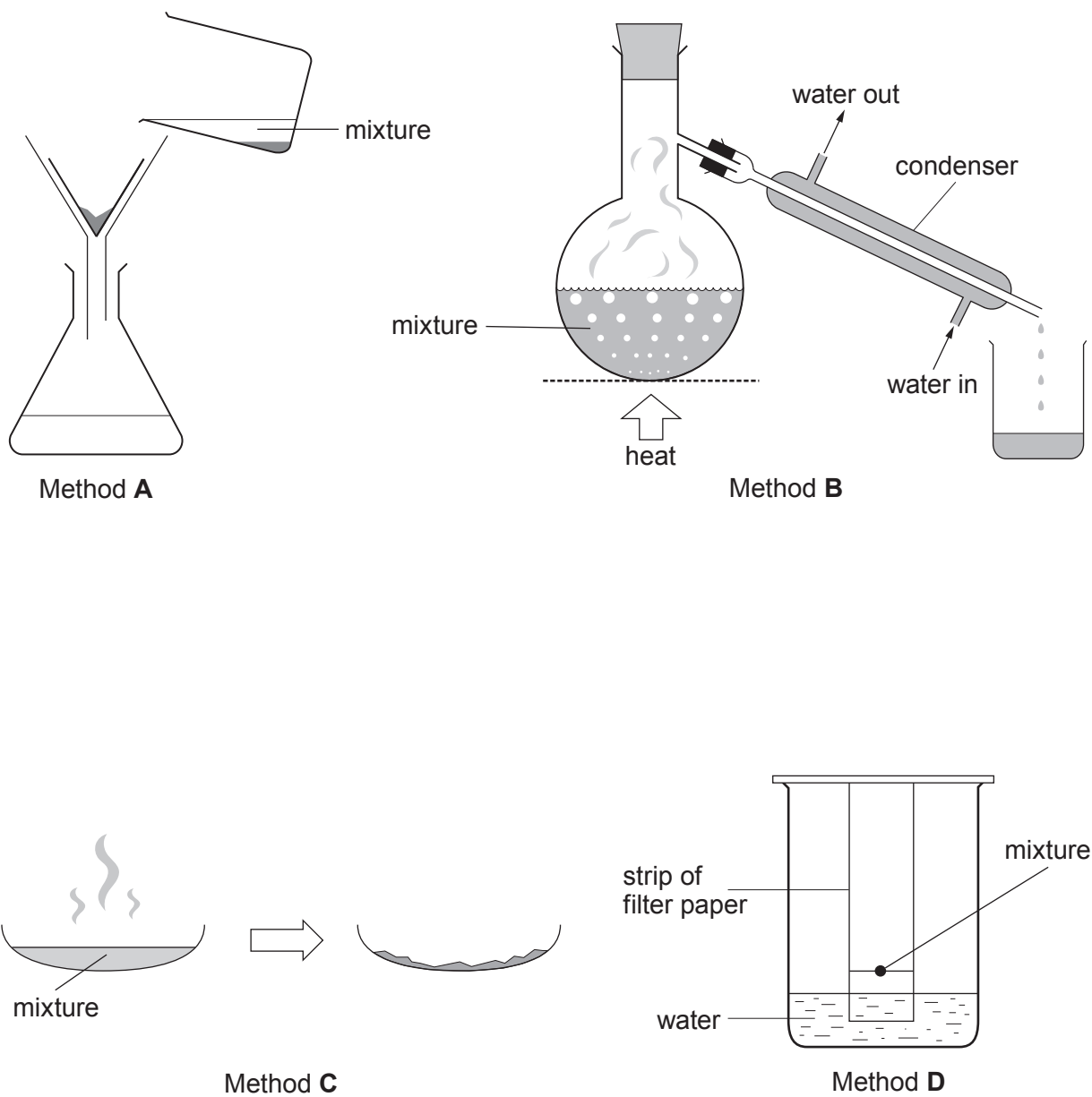
One line has been drawn for you. [2]

Metal	Colour seen
	white
lithium	red
sodium	green
barium	blue
	yellow



Answer **all** questions.

1. (a) The diagrams show four methods, **A**, **B**, **C** and **D**, used to separate different mixtures.



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(i) Choose from the box the names of methods **B** and **D**. [2]

distillation	chromatography	filtration	evaporation	boiling
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Method **B**

Method **D**

(ii) Give the letter, **A**, **B**, **C** or **D**, of the method used to [3]

remove sand from water

obtain pure water from sea water

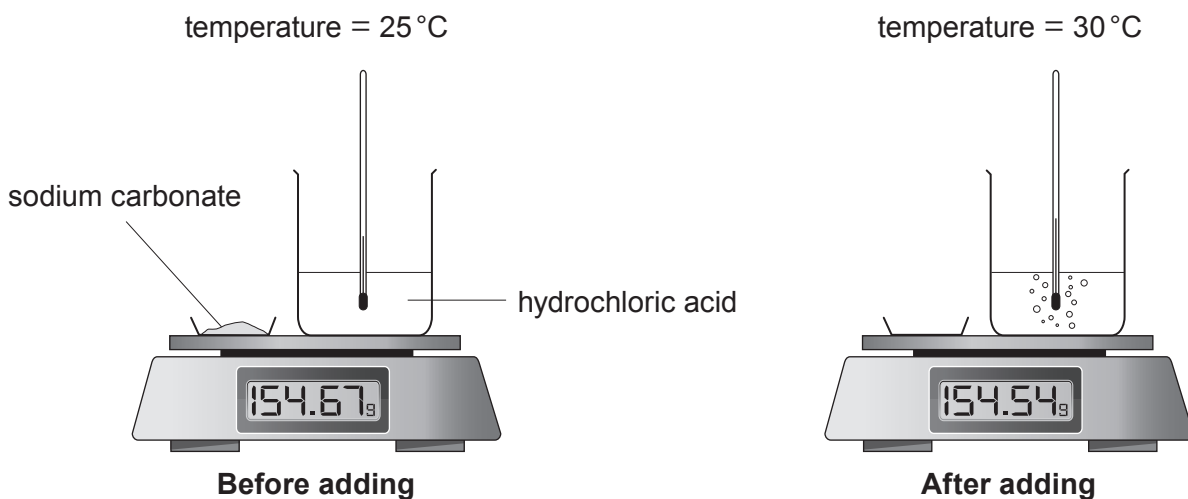
separate red and yellow dyes

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- (b) Sodium carbonate reacts with dilute hydrochloric acid forming sodium chloride, water and carbon dioxide.

The diagrams show the apparatus before and after sodium carbonate is added to hydrochloric acid.



Tick (✓) **two** observations that show a chemical reaction is taking place.

[2]

The solid stays the same

☐

A gas is formed

☐

A temperature change occurs

☐

The mass of the beaker and contents stays the same

☐